

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461248.7938 +/- 0.009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.046
Transit depth (Rp/Rs)^2	2.52 +/- 1.47 [%]
Orbital Inclination (inc)	88.73 +/- 2.93
Airmass coefficient 1 (a1)	0.061 +/- 0.047
Airmass coefficient 2 (a2)	2.16 +/- 1.57
Scatter in the residuals of the lightcurve fit is	94.96 %
Best Comparison Star	None
Optimal Aperture	5.06
Optimal Annulus	6.75
Transit Duration (day)	0.082 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.046 vs 0.1326 (deviation 0.37σ)
Duration fit vs published	0.082 d vs 0.0737 d (deviation 0.33σ)
Mid-time O–C	-20.2 min
Depth SNR	2.2
Residual scatter	94.96 %

Light curve

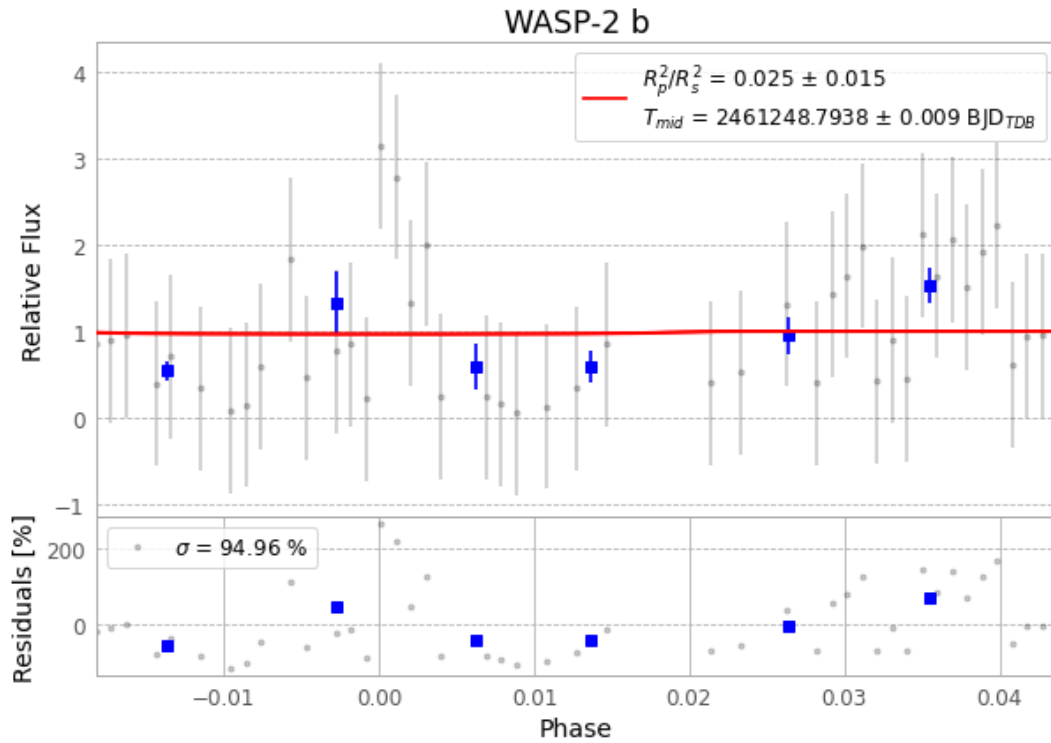


Figure 1 — FinalLightCurve_WASP-2 b_2026-07-26.png (SHA-256 7fee7a95fbca6823...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 40], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 45ab500f692f84ed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), WASP-2260727055716.FITS → WASP-2260727090916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-260727.log (e77d2beab952...), FinalLightCurve_WASP-2 b_2026-07-26.png (7fee7a95fbca...), FinalLightCurve_WASP-2 b_2026-07-26.csv (a5fdaf2a1bec...)

Compute resources

Wall time 184.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 23:27Z.